

CLIQUE PARTITIONS OF CHORDAL GRAPHS WITH LINEAR ERROR

ANONYMOUS

ABSTRACT. We prove that the edges of every chordal graph on n vertices can be partitioned into $n^2/6 + O(n)$ complete subgraphs. This resolves a problem of Erdős, Ordman, and Zalcstein. In fact, for all sufficiently large n , the maximum clique-partition number among n -vertex chordal graphs is

$$\left\lfloor \frac{n(n+1)}{6} \right\rfloor.$$

The proof combines a weighted fractional packing of triangles and copies of K_4 with a quantitative stability argument. Near the extremal complete-split graphs we construct a partition using only edges and triangles, with a strict saving for every structural defect. Far from that family, a finite-family packing transfer absorbs its subquadratic rounding error. A single-vertex symmetrization path and a first-entry argument connect the two regimes.

1. INTRODUCTION

A *clique partition* of a finite simple graph G is a collection of complete subgraphs whose edge sets partition $E(G)$. We write $\text{cp}(G)$ for the minimum number of parts in such a partition. Erdős, Ordman, and Zalcstein asked whether every chordal graph on n vertices satisfies

$$\text{cp}(G) \leq \frac{n^2}{6} + O(n). \quad (1.1)$$

Their complete-split examples show that the coefficient $1/6$ cannot be improved [EOZ93]. The question is listed as Erdős Problem 81 [Blo26].

Recall that a graph is chordal if it contains no induced cycle of length at least four. Split graphs, whose vertices can be divided into a clique and an independent set, form an important subclass. Chen, Erdős, and Ordman proved the bound $3n^2/16 + O(n)$ for split graphs [CEO94]. Traverso determined the exact extremum of a fractional triangle-cover functional over chordal graphs, with a complete-split extremizer and a clone-copy reduction [TG26a]. Combined with the standard dense packing transfer, that result already gives the first-order estimate $n^2/6 + o(n^2)$ for chordal graphs. Traverso

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subsequently proved the linear-error bound for split graphs [TG26b], while Cipollini independently obtained $n^2/6 + o(n^2)$ for all chordal graphs [Cip26]. The remaining issue addressed here is the linear-error conclusion for general chordal graphs.

Our main result is stronger than (1.1).

Theorem 1.1. *There is an integer N such that, for every $n \geq N$,*

$$\max\{\text{cp}(G) : G \text{ is chordal and } |V(G)| = n\} = M(n) := \left\lfloor \frac{n(n+1)}{6} \right\rfloor.$$

Corollary 1.2. *There is an absolute constant C such that every chordal graph G on n vertices satisfies*

$$\text{cp}(G) \leq \frac{n^2}{6} + Cn.$$

The proof has two distinct parts. We first introduce a mixed packing functional in which a triangle has gain 2 and a copy of K_4 has gain 5. The K_4 term is structurally useful rather than cosmetic. For the triangle-only fractional functional, complete graphs remain on the competing $n^2/6$ scale; allowing K_4 with its natural gain $\binom{4}{2} - 1 = 5$ lowers that branch by a fixed quadratic amount. This isolates the complete-split branch needed by the stability argument. The corresponding fractional functional is well behaved under vertex copying. Repeated copies lead monotonically to a complete-split graph. This gives the correct global quadratic bound, but a general packing-transfer theorem loses $o(n^2)$ and therefore cannot by itself prove (1.1).

The second part repairs precisely that loss near the extremal family. Starting from a clique of order close to $n/3$, we regularize the choice of root clique without changing the graph. A balanced Vizing edge-colouring and the Häggkvist–Janssen list-edge-colouring theorem then produce an integral triangle packing with a fixed saving for both types of defect: edges outside the root and missing edges between the root and its complement. This yields a local stability inequality in the actual labelled edit distance. Finally, a single-vertex version of the copying argument forces every graph with nearly extremal fractional value into that local neighbourhood. The first path vertex entering the neighbourhood would otherwise satisfy incompatible upper and lower bounds on its edit distance.

The constants used below are intentionally conservative. Put

$$\rho = 10^{-12}, \quad \eta = 10^{-30}. \tag{1.2}$$

For the packing-transfer theorem stated in theorem 2.3, choose a threshold $T(\varepsilon)$ that works uniformly for all graphs, and set

$$N = \max\{10^{32}, T(\eta/2)\}. \tag{1.3}$$

No numerical value for $T(\eta/2)$ is asserted.

2. PRELIMINARIES AND EXTERNAL INPUTS

All graphs in the paper are finite and simple. For $r \geq 2$, let $\text{cp}_{\leq r}(G)$ be the minimum number of parts in a clique partition of G in which every part has at most r vertices. Thus

$$\text{cp}(G) \leq \text{cp}_{\leq r}(G).$$

We allow copies of K_2 as parts. We write $e(G) = |E(G)|$, $\Delta(G)$ for maximum degree, and $\omega(G)$ for clique number.

We use the following three published results. Their quantifiers are recorded explicitly because the uniformity in the third result is important.

Theorem 2.1 (Vizing [Viz64]). *Every finite simple graph H has a proper edge-colouring with at most $\Delta(H) + 1$ colours.*

Theorem 2.2 (Häggkvist–Janssen [HJ97]). *The list-chromatic index of K_p is at most p . Equivalently, if every edge of K_p is assigned a list of at least p colours, then the edges admit a proper colouring in which each edge receives a colour from its own list.*

For a graph G , consider edge-disjoint packings by triangles and copies of K_4 . Assign gain 2 to a triangle and gain 5 to a copy of K_4 . Let $W_4(G)$ be the maximum total gain of an integral packing and let $W_4^*(G)$ be the optimum of the fractional relaxation. Explicitly, if $\mathcal{T}(G)$ and $\mathcal{K}_4(G)$ denote the triangles and copies of K_4 in G , then

$$\begin{aligned} W_4^*(G) = \max \quad & 2 \sum_{T \in \mathcal{T}(G)} x_T + 5 \sum_{K \in \mathcal{K}_4(G)} z_K, \\ \text{subject to} \quad & x_T, z_K \geq 0, \\ & \sum_{T \ni e} x_T + \sum_{K \ni e} z_K \leq 1 \quad \text{for every } e \in E(G). \end{aligned} \tag{2.1}$$

The gains are exactly the numbers of parts saved by replacing constituent edges with one clique: $\binom{3}{2} - 1 = 2$ and $\binom{4}{2} - 1 = 5$. Thus the family and weight vector in the notation of the finite-family transfer theorem are $\{K_3, K_4\}$ and $(2, 5)$.

Theorem 2.3 (finite-family packing transfer). *For every $\varepsilon > 0$ there is an integer $T(\varepsilon)$ such that every graph G with $|V(G)| \geq T(\varepsilon)$ satisfies*

$$0 \leq W_4^*(G) - W_4(G) \leq \varepsilon |V(G)|^2.$$

The threshold is uniform over all graphs of the given order.

This is the specialization of the weighted finite-family theorem [RUW21, Theorem 3.4] to $\{K_3, K_4\}$ and gains 2, 5. The graph-independent threshold used here is the uniformity made explicit in the proof of [RUW21, Theorem 3.6].

We shall use standard properties of chordal graphs: induced subgraphs are chordal; chordal graphs admit perfect elimination orderings; adding a vertex

whose neighbourhood is a clique preserves chordality; and every noncomplete chordal graph has two nonadjacent simplicial vertices [Dir61, BP93]. We also need the following convenient form of the elimination-ordering property.

Lemma 2.4. *If P is a clique in a chordal graph G , then G has a perfect elimination ordering whose final vertices are precisely the vertices of P .*

Proof. As long as a vertex remains outside P , remove a simplicial vertex outside P . Such a vertex exists: if the current graph is complete, every outside vertex is simplicial; otherwise, two nonadjacent simplicial vertices exist and they cannot both lie in the clique P . The induced graph left after each removal is chordal. Appending the vertices of P in any order completes the desired ordering. $\square$

The fractional mixed packing has the dual formulation

$$W_4^*(G) = \min \left\{ \sum_{e \in E(G)} y_e : \begin{array}{ll} y_e \geq 0, & \\ \sum_{e \in E(T)} y_e \geq 2 & \text{for every triangle } T, \\ \sum_{e \in E(K)} y_e \geq 5 & \text{for every } K \cong K_4 \end{array} \right\}. \quad (2.2)$$

Both primal and dual optima are attained. For example, a dual coordinate larger than 5 can be replaced by 5, so the dual minimum may be taken over a compact box.

Define

$$\Phi(G) = e(G) - W_4^*(G), \quad Q(n) = \frac{(2n+1)^2}{24}. \quad (2.3)$$

Since the gain of a clique is the number of edges saved when it replaces its constituent K_2 parts, we have the exact identity

$$\text{cp}_{\leq 4}(G) = e(G) - W_4(G) = \Phi(G) + W_4^*(G) - W_4(G). \quad (2.4)$$

Moreover,

$$\Phi(G) \leq \text{cp}_{\leq 3}(G), \quad (2.5)$$

because every integral triangle packing is an admissible mixed fractional packing. Notice that (2.5) does not assert $\Phi(G) \leq \text{cp}(G)$.

Finally,

$$\lfloor Q(n) \rfloor = \left\lfloor \frac{n(n+1)}{6} \right\rfloor = M(n). \quad (2.6)$$

Indeed, $Q(n) = n(n+1)/6 + 1/24$, while the fractional part of $n(n+1)/6$ is either 0 or $1/3$.

3. AN INTEGRAL CONSTRUCTION AROUND A ROOT CLIQUE

Let P be a clique in a chordal graph G , and introduce the data

$$p = |P|, \quad H = G - P, \quad q = |V(H)|, \quad m = e(H), \quad w = \omega(H). \quad (3.1)$$

Write

$$A = pq - e_G(P, V(H)), \quad D = \max_{x \in P} |V(H) \setminus N_G(x)|. \quad (3.2)$$

Thus A is the total number of missing root–outside incidences and D is the largest missing column; here $e_G(X, Y)$ denotes the number of edges with one endpoint in each of the disjoint sets X and Y .

Lemma 3.1 (terminal construction). *Assume $p \geq 1$, and put*

$$c = \max\{p, \Delta(H) + 1\}, \quad t = \left\lceil \frac{m}{c} \right\rceil.$$

If

$$q - 2D - 4t \geq p, \tag{3.3}$$

then

$$\text{cp}_{\leq 3}(G) \leq pq - \binom{p}{2} + \left(1 - \frac{2p}{c}\right)m - \left(1 - \frac{2(w-1)}{p}\right)A. \tag{3.4}$$

The clique P need not be maximum.

Proof. By theorem 2.1, the edges of H have a proper colouring with c colours. Choose such a colouring for which the sum of the squares of the colour-class sizes is minimum. Its classes differ in size by at most one. Indeed, if two classes differed by at least two, their union would contain an alternating path with one more edge of the larger colour; interchanging the two colours on that path would decrease the sum of squares. Consequently each class has at most t edges.

Retain the p largest classes. Together they contain at least pm/c edges. We assign these classes bijectively to the vertices of P . An outside edge uv assigned to $x \in P$ is retained when x is adjacent to both u and v ; it then yields the triangle uvx .

Choose a perfect elimination ordering ending in P , as in lemma 2.4, and orient every edge of H from its earlier to its later endpoint. If uv is oriented from u to v , then

$$N_G(u) \cap P \subseteq N_G(v) \cap P,$$

because the later neighbours of u form a clique. Hence the number of root vertices that are invalid hosts for uv is exactly $a_u = p - |N_G(u) \cap P|$. The vertex u has at most $w - 1$ later neighbours in H , and therefore the total number of invalid-host incidences over all outside edges is at most

$$(w - 1) \sum_{u \in V(H)} a_u = (w - 1)A.$$

This remains an upper bound after only the selected colour classes are kept.

Choose the bijection from the selected classes to P uniformly at random. Each invalid host is assigned to its class with probability $1/p$, so some bijection retains at least

$$f \geq \frac{pm}{c} - \frac{(w - 1)A}{p} \tag{3.5}$$

outside edges. Use the corresponding f triangles. Since every selected class is a matching of size at most t , each root vertex is incident with at most $2t$ outside spokes already used by these triangles.

It remains to cover the edges within P . For each $xy \in E(G[P])$, let $L(xy)$ be the set of outside vertices z such that xyz is a triangle and neither xz nor yz has already been used. The union bound gives

$$|L(xy)| \geq q - 2D - 4t \geq p.$$

Apply theorem 2.2 to the complete graph on P with these actual lists. Its proper list-edge-colouring assigns a host $z \in L(xy)$ to every root edge xy . The resulting root triangles are pairwise edge-disjoint and are also edge-disjoint from the f outside-edge triangles.

Use K_2 parts for every crossing or outside edge not yet covered. The exact number of parts is

$$pq - \binom{p}{2} + m - A - 2f.$$

Substituting (3.5) proves (3.4). $\square$

We shall repeatedly use the following immediate consequence.

Corollary 3.2. *Under the hypotheses of lemma 3.1, if in addition*

$$c \leq \frac{9p}{5} \quad \text{and} \quad \frac{2(w-1)}{p} \leq \frac{1}{2},$$

then

$$\text{cp}_{\leq 3}(G) \leq pq - \binom{p}{2} - \frac{m}{9} - \frac{A}{2}. \quad (3.6)$$

Proof. In (3.4), the coefficient of m is at most $-1/9$ and the coefficient of A is at most $-1/2$. $\square$

4. STRICT REGULARIZATION OF THE ROOT

The construction in lemma 3.1 becomes useful only when both coefficients in (3.4) are bounded away from zero. The next elementary observation controls vertices that are not complete to the root.

Lemma 4.1. *With the notation of (3.1)–(3.2), suppose $u \in V(H)$ is nonadjacent to some $x \in P$. Then*

$$d_H(u) \leq D + w - 2.$$

Proof. The common neighbours of two nonadjacent vertices in a chordal graph form a clique: two nonadjacent common neighbours would induce a 4-cycle. Thus $N_H(u) \cap N_G(x)$, together with u , is a clique in H , and the intersection has at most $w - 1$ vertices. Every remaining neighbour of u in H lies in $V(H) \setminus N_G(x)$. This set has at most D vertices and contains u itself, so it contains at most $D - 1$ neighbours of u . Adding the two estimates proves the claim. $\square$

Lemma 4.2 (strict root regularization). *Suppose that a chordal graph G contains a clique P whose data satisfy*

$$p \geq 128, \quad |q - 2p| \leq \frac{p}{64}, \quad m + A \leq \frac{p^2}{65536}. \quad (4.1)$$

Then G contains a clique P' —not necessarily a maximum clique—such that, for the corresponding data p', q', m', A' ,

$$\text{cp}_{\leq 3}(G) \leq p'q' - \binom{p'}{2} - \frac{m'}{9} - \frac{A'}{2}. \quad (4.2)$$

Proof. First demote every root vertex whose original missing column has more than $p/64$ entries. Let R be the set of demoted vertices, let $r = |R|$, and put $P_0 = P \setminus R$. This changes only the designation of the root: no edge is edited and no clique-partition part is chosen. From (4.1),

$$r \leq \frac{64A}{p} \leq \frac{p}{1024}, \quad p_0 = p - r \geq \frac{99p}{100}, \quad q_0 = q + r, \quad (4.3)$$

and the largest missing column of the retained root satisfies $D_0 \leq p/64$.

Let $H_0 = G - P_0$. Since R was contained in the original clique, its outside edge count is exactly

$$m_0 = m + e_G(R, V(H)) + \binom{r}{2}.$$

Using $q \leq 129p/64$ and (4.3), we obtain

$$\begin{aligned} m_0 &\leq \left(\frac{1}{65536} + \frac{129}{64 \cdot 1024} + \frac{1}{2 \cdot 1024^2} \right) p^2 \\ &= \frac{4161}{2097152} p^2 < \frac{p^2}{400}. \end{aligned} \quad (4.4)$$

If $w_0 = \omega(H_0)$, then $\binom{w_0}{2} \leq m_0$, and hence

$$w_0 - 1 \leq \sqrt{2m_0} < \frac{p}{10}. \quad (4.5)$$

Define the fixed defect bound

$$\overline{D} = \max \left\{ D_0, q_0 - \frac{7p_0}{4} \right\}. \quad (4.6)$$

Equations (4.1) and (4.3) give

$$\overline{D} \leq \max \left\{ \frac{p}{64}, \left(\frac{17}{64} + \frac{11}{4096} \right) p \right\} = \frac{1099}{4096} p < \frac{p}{3}. \quad (4.7)$$

Starting from P_0 , promote an outside vertex whenever its degree in the current outside graph is at least $7/4$ times the current root order. Every promotion is legal. Indeed, by lemma 4.1, a vertex not complete to the current root has outside degree at most

$$\overline{D} + w_0 - 2 < \frac{13p}{30},$$

far below $7p_0/4$. Thus every promoted vertex is complete to the current root, which consequently remains a clique.

The bound $\overline{D}$ remains valid throughout the procedure. If u is promoted when the outside and root orders are q_i and p_i , respectively, then its missing column after promotion has size at most

$$q_i - 1 - d_{H_i}(u) \leq q_i - \frac{7p_i}{4} - 1 \leq q_0 - \frac{7p_0}{4} \leq \overline{D}.$$

The old missing columns can only shrink, and the outside clique number cannot increase.

The edges removed from the outside graph in different promotions are disjointly accounted for. If h vertices are promoted, then

$$h \leq \frac{4m_0}{7p_0} \leq \frac{p}{693}. \quad (4.8)$$

Let P' and H' be the root and outside graph at termination. Then $\Delta(H') < 7p'/4$, and therefore

$$c' := \max\{p', \Delta(H') + 1\} \leq \left\lceil \frac{7p'}{4} \right\rceil \leq \frac{9p'}{5}. \quad (4.9)$$

The last inequality holds for every integer $p' \geq 15$, whereas here $p' \geq p_0 \geq 127$. Also, by (4.5),

$$\frac{2(w' - 1)}{p'} < \frac{20}{99} < \frac{1}{2}. \quad (4.10)$$

Put $t' = \lceil m'/c' \rceil$. Since $m' \leq m_0$ and $c' \geq p_0$,

$$t' \leq \left\lceil \frac{m_0}{p_0} \right\rceil \leq \frac{p}{396} + 1. \quad (4.11)$$

Finally, each promotion decreases $q - p$ by two, so

$$\begin{aligned} q' - p' - 2D' - 4t' &\geq \left(\frac{63}{64} - \frac{2}{693} - \frac{2}{3} - \frac{1}{99} \right) p - 4 \\ &= \frac{4505}{14784} p - 4 > \frac{p}{4} - 4 \geq 0. \end{aligned} \quad (4.12)$$

Thus the host condition (3.3) holds for P' . Apply corollary 3.2 using (4.9) and (4.10); this gives (4.2). $\square$

Remark 4.3. The promotion threshold $7p/4$ is what leaves a fixed negative coefficient on m' in (4.2). A threshold of $2p$ would permit that coefficient to vanish. The exact formula for m_0 also shows that every edge created in the outside graph by demoting a root vertex has been charged.

5. QUANTITATIVE LOCAL STABILITY

Fix n , put $a = \lfloor n/3 \rfloor$, and, for each $A_0 \subseteq V(G)$ with $|A_0| = a$, let S_{A_0} be the complete-split graph on $V(G)$ with clique side A_0 . Thus two vertices

are adjacent in S_{A_0} if and only if at least one of them lies in A_0 . Define the labelled edit distance

$$d_{\text{sp}}(G) = \min_{|A_0|=a} |E(G) \triangle E(S_{A_0})|. \quad (5.1)$$

A single operation that replaces the neighbourhood of one vertex by the neighbourhood of a nonadjacent vertex changes at most $n - 2$ unordered pairs. Consequently, d_{sp} changes by at most $n - 2$ in one such operation.

Lemma 5.1. *Let G be chordal of order $n \geq 1000$. If*

$$d_{\text{sp}}(G) \leq \rho n^2,$$

then G has a clique satisfying the hypotheses of lemma 4.2.

Proof. Choose A_0 in (5.1) with edit count $E \leq \rho n^2$. Let P be a maximum clique of the chordal graph $G[A_0]$, write $p = |P|$, and put $u = a - p$.

A chordal graph on a vertices with clique number p has at most

$$(p - 1)a - \binom{p}{2} \quad (5.2)$$

edges. To see this, sum the number of later neighbours in a perfect elimination ordering, bounding the successive terms by $\min\{p - 1, a - i\}$. The number of pairs missing from $G[A_0]$ is therefore at least $\binom{u+1}{2}$. These pairs are counted by E , and hence

$$u \leq \sqrt{2E} \leq \sqrt{2\rho} n. \quad (5.3)$$

Since $p = a - u$ and $q = n - p$, for $n \geq 1000$ we have

$$p \geq \frac{n}{3} - 1 - \sqrt{2\rho} n \geq \frac{33n}{100} \geq 128, \quad (5.4)$$

$$p \leq \frac{n}{3}, \quad 0 \leq q - 2p = n - 3p \leq 3 + 3\sqrt{2\rho} n \leq \frac{p}{64}. \quad (5.5)$$

The edges in the old independent side and the nonedges between the two old sides contribute at most E defects altogether. Every newly relevant pair involving $A_0 \setminus P$ contributes at most un . Thus the outside-edge count and missing-incidence count for the root P satisfy

$$\begin{aligned} m + A &\leq E + un \leq (\rho + \sqrt{2\rho})n^2 \\ &\leq \left(10^{-12} + \frac{3}{2 \cdot 10^6}\right)n^2 < \frac{(33/100)^2 n^2}{65536} \leq \frac{p^2}{65536}. \end{aligned} \quad (5.6)$$

Together with (5.4)–(5.5), this is exactly (4.1). $\square$

Lemma 5.2 (local stability). *Under the hypotheses of lemma 5.1, put*

$$\mathcal{D} = Q(n) - \Phi(G).$$

Then $\mathcal{D} \geq 0$ and

$$d_{\text{sp}}(G) \leq 9\mathcal{D} + n\sqrt{\frac{2\mathcal{D}}{3}} + n, \quad (5.7)$$

$$\text{cp}_{\leq 3}(G) \leq M(n). \quad (5.8)$$

Proof. Apply lemma 4.2, and use primes for the resulting root data. By (2.5) and (4.2),

$$\mathcal{D} \geq Q(n) - \left(p'q' - \binom{p'}{2} \right) + \frac{m'}{9} + \frac{A'}{2}. \quad (5.9)$$

The first difference on the right has the exact form

$$Q(n) - \left(p'(n - p') - \binom{p'}{2} \right) = \frac{(6p' - 2n - 1)^2}{24}. \quad (5.10)$$

It follows that

$$m' + A' \leq 9\mathcal{D}, \quad \left| p' - \frac{2n+1}{6} \right| \leq \sqrt{\frac{2\mathcal{D}}{3}}. \quad (5.11)$$

In particular, $\mathcal{D} \geq 0$.

The graph G differs from the complete-split graph with clique side P' on exactly $m' + A'$ pairs. Changing that clique side to have order $\lfloor n/3 \rfloor$ requires at most $|p' - \lfloor n/3 \rfloor|$ vertex-role changes, each costing at most n edge edits. Since

$$\left| \frac{2n+1}{6} - \left\lfloor \frac{n}{3} \right\rfloor \right| < 1,$$

equation (5.7) follows from (5.11).

Finally, (4.2) and (5.10) give $\text{cp}_{\leq 3}(G) \leq Q(n)$. The left side is an integer, so (2.6) yields (5.8). $\square$

6. A MONOTONE PATH OF SINGLE-VERTEX COPIES

The cover-pullback copy inequality, its discrete-convex lift to clone classes, and the complete-split terminal reduction follow the strategy of Traverso [TG26a, Sections 3–5] for the triangle-cover functional. We give the mixed-functional adaptation because both triangle and K_4 constraints must be transported. The additional feature needed later is a fine path recording every single-vertex copy, so that a first-entry argument can control the moment the path crosses a fixed edit-distance ball.

Let u and v be nonadjacent vertices of a graph G . Write $G_{u \rightarrow v}$ for the graph obtained by deleting the edges incident with u and then making $N_{G_{u \rightarrow v}}(u) = N_G(v)$. In particular, u and v remain nonadjacent.

Lemma 6.1 (copy inequality). *For every graph G and every nonadjacent pair u, v ,*

$$\Phi(G_{u \rightarrow v}) + \Phi(G_{v \rightarrow u}) \geq 2\Phi(G). \quad (6.1)$$

Proof. Choose an optimal dual cover y in (2.2). To obtain a cover of $G_{u \rightarrow v}$, keep every unaffected edge weight and give each new edge ux the old weight y_{vx} . Every triangle or K_4 containing u is then a copy of one containing v in G , while no clique contains both u and v . Thus all dual constraints remain valid.

Let s_u and s_v be the sums of the y -weights on edges incident with u and v . The copied covers of $G_{u \rightarrow v}$ and $G_{v \rightarrow u}$ have respective weights

$$W_4^*(G) - s_u + s_v \quad \text{and} \quad W_4^*(G) - s_v + s_u.$$

They therefore give

$$W_4^*(G_{u \rightarrow v}) + W_4^*(G_{v \rightarrow u}) \leq 2W_4^*(G).$$

On the other hand, the two copied edge counts sum to $2e(G)$. Subtracting the preceding inequality from this identity gives (6.1). $\square$

Vertices with the same open neighbourhood are called *false twins*. Every false-twin class is independent. Suppose A_1 and A_2 are distinct false-twin classes consisting of simplicial vertices, with no edge between the two classes. Keep the graph outside $A_1 \cup A_2$ fixed. If $r = |A_1| + |A_2|$, let H_j be the graph in which j of these r vertices have neighbourhood $N_G(A_1)$ and the other $r - j$ have neighbourhood $N_G(A_2)$. For fixed j , different choices of the j labelled vertices give isomorphic graphs, via a permutation supported on $A_1 \cup A_2$; hence $\Phi(H_j)$ depends only on j . Both neighbourhoods are cliques, so every H_j is chordal. For $0 < j < r$, lemma 6.1 gives

$$\Phi(H_{j-1}) + \Phi(H_{j+1}) \geq 2\Phi(H_j). \quad (6.2)$$

Thus $j \mapsto \Phi(H_j)$ is discretely convex. At any interior index, at least one adjacent value is no smaller. Once such a direction is chosen, the successive differences in that direction remain nonnegative by (6.2). We may therefore move one vertex at a time to the corresponding endpoint without ever decreasing Φ .

At the endpoint the two selected false-twin classes have merged. No class outside their union splits: all of its vertices have the same adjacency to a representative of the target class and consequently acquire the same adjacency to every copied vertex. The number of false-twin classes has strictly decreased.

We next identify the graphs at which this procedure stops; this is the same terminal family appearing in [TG26a, Section 5].

Lemma 6.2 (terminal characterization). *Let G be chordal. If every two nonadjacent simplicial vertices of G have the same open neighbourhood, then G is complete-split; that is,*

$$G \cong K_k \vee \overline{K}_{n-k}$$

for some $0 \leq k \leq n$.

Proof. Suppose first that G is disconnected and has an edge. Choose a nonisolated simplicial vertex in a nontrivial component and a simplicial vertex in another component. Their neighbourhoods are disjoint and at least one is nonempty, contrary to the hypothesis. Hence a disconnected terminal graph is edgeless. Complete graphs are also of the required form.

It remains to consider a connected, noncomplete graph. Take a clique tree of G with at least two maximal-clique bags. A vertex private to a leaf bag is

simplicial and occurs in no other bag. Private vertices chosen from distinct leaves are nonadjacent, and hence they have a common open neighbourhood, say K .

Each leaf has exactly one private vertex. Otherwise, the neighbourhood of a private vertex in that leaf would contain another private vertex from the same leaf, whereas the neighbourhood of a private vertex in a different leaf would not. It follows that every leaf bag is K together with its unique private vertex.

Every member of K occurs in every leaf bag. By the running-intersection property, it then occurs in every bag, because every node of a finite tree lies on a path between two leaves. Thus K is complete and universal to $V(G) \setminus K$. Each leaf-private vertex is isolated in $G - K$.

If $G - K$ had a nontrivial component, choose a nonisolated simplicial vertex of that chordal component. Its neighbourhood in G is a clique, so it is also simplicial in G . It is nonadjacent to a leaf-private vertex, but its neighbourhood properly contains K , again a contradiction. Therefore $G - K$ is independent, as required. $\square$

Proposition 6.3 (fine monotone symmetrization). *Every chordal graph G of order n admits a sequence*

$$G = G_0, G_1, \dots, G_L = S_{k,n-k}$$

of chordal graphs such that

$$\Phi(G_{i+1}) \geq \Phi(G_i) \quad \text{and} \quad |E(G_{i+1}) \triangle E(G_i)| \leq n - 2$$

for every i . One may take $L \leq n(n - 1)$.

Proof. If the condition in lemma 6.2 fails, choose two nonadjacent simplicial vertices with different neighbourhoods and take their false-twin classes. The preceding discrete-convexity argument merges those classes by a sequence of nondecreasing single-vertex copies. Each copied vertex changes at most $n - 2$ incident pairs. Repeat. A full-class merge strictly decreases the number of false-twin classes, so there are at most $n - 1$ merges, each requiring at most n copies. The terminal graph is complete-split by lemma 6.2. $\square$

The single-vertex resolution in proposition 6.3 is essential: the proof below controls the first point at which the path crosses the boundary of a fixed edit-distance neighbourhood.

7. THE MIXED FUNCTIONAL ON COMPLETE-SPLIT GRAPHS

Write

$$S_{k,\ell} = K_k \vee \overline{K}_\ell, \quad k + \ell = n.$$

Lemma 7.1. *If $k \geq 4$ and $\ell \geq 1$, then*

$$\Phi(S_{k,\ell}) = \max \left\{ k\ell - \binom{k}{2}, \frac{2k\ell - \binom{k}{2}}{3}, \frac{k\ell + \binom{k}{2}}{6} \right\}. \quad (7.1)$$

Proof. Average an optimal dual cover over all automorphisms of $S_{k,\ell}$. We may therefore assign a common weight x to each edge in the clique side and a common weight y to each spoke. The triangle and K_4 constraints in (2.2) reduce to

$$x \geq \frac{5}{6}, \quad y \geq 0, \quad x + 2y \geq 2, \quad x + y \geq \frac{5}{3}. \quad (7.2)$$

Here $x \geq 5/6$ subsumes the root-triangle constraint $x \geq 2/3$.

The lower boundary of (7.2) has vertices

$$(2, 0), \quad \left(\frac{4}{3}, \frac{1}{3}\right), \quad \left(\frac{5}{6}, \frac{5}{6}\right).$$

The dual objective is $\binom{k}{2}x + k\ell y$, so its minimum is attained at one of these vertices. Subtracting the three possible values from $e(S_{k,\ell}) = \binom{k}{2} + k\ell$ gives (7.1). $\square$

The three expressions in (7.1), maximized over the real variable k , are bounded respectively by

$$Q(n), \quad \frac{(4n+1)^2}{120}, \quad qquad \frac{n(n-1)}{12}. \quad (7.3)$$

The first bound is the square identity (5.10); the other two follow by completing the square and by $e(S_{k,\ell}) \leq \binom{n}{2}$. If $\ell = 0$ and $k \geq 4$, then $\Phi(K_k) = e(K_k)/6$: the constant dual weight $5/6$ is optimal, as witnessed by the uniform fractional packing of all copies of K_4 . If $k \leq 3$, then $\Phi(S_{k,n-k}) \leq e(S_{k,n-k}) \leq 3n$.

For $n \geq 100$, every branch except the first in (7.1), including the preceding degenerate cases, is strictly smaller than

$$\frac{n^2}{6} - \frac{n^2}{40}. \quad (7.4)$$

For the second branch this is equivalent to $n^2 > 8n + 1$, while $3n < n^2/6 - n^2/40$ follows from $17n > 360$; the third branch and the complete graph are smaller still.

Corollary 7.2. *Every chordal graph G of order $n \geq 100$ satisfies*

$$\Phi(G) \leq Q(n).$$

Proof. Apply proposition 6.3. The functional does not decrease along the path, and (7.3) bounds its complete-split endpoint by $Q(n)$. $\square$

We shall also need a quantitative statement about a nearly extremal terminal graph.

Lemma 7.3. *Let $n \geq 100$, and suppose that a complete-split graph S of order n satisfies*

$$\Phi(S) \geq \frac{n^2}{6} - \eta n^2.$$

If $\mathcal{D} = Q(n) - \Phi(S)$, then the first branch in (7.1) gives the value of $\Phi(S)$ and

$$d_{\text{sp}}(S) \leq n\sqrt{\frac{2\mathcal{D}}{3}} + n. \quad (7.5)$$

Proof. Since $\eta = 10^{-30} < 1/40$, the branch separation (7.4) forces the first branch. If its root has order k , the square identity (5.10) gives

$$\left| k - \frac{2n+1}{6} \right| \leq \sqrt{\frac{2\mathcal{D}}{3}}.$$

Changing the root order to $\lfloor n/3 \rfloor$ costs at most n edits per vertex-role change. The offset between $(2n+1)/6$ and $\lfloor n/3 \rfloor$ is less than one, proving (7.5). $\square$

8. GLOBAL STABILITY BY FIRST ENTRY

Theorem 8.1 (fixed-neighbourhood stability). *Let G be chordal of order $n \geq 10^{32}$. If*

$$\Phi(G) \geq \frac{n^2}{6} - \eta n^2, \quad (8.1)$$

then

$$d_{\text{sp}}(G) < \frac{\rho n^2}{2}. \quad (8.2)$$

Proof. Take the path $G = G_0, \dots, G_L$ from proposition 6.3. Its functional values are no smaller than $\Phi(G)$. Therefore, for every graph H on the path,

$$Q(n) - \Phi(H) \leq \eta n^2 + \frac{n}{6} + \frac{1}{24} \leq 2\eta n^2, \quad (8.3)$$

where the last inequality follows from $n \geq 10^{32}$. By lemma 7.3, the endpoint satisfies

$$\frac{d_{\text{sp}}(G_L)}{n^2} \leq \sqrt{\frac{4\eta}{3}} + \frac{1}{n} < \frac{\rho}{4}.$$

Suppose, contrary to (8.2), that $d_{\text{sp}}(G_0) \geq \rho n^2/2$. Let i be the first path index with $d_{\text{sp}}(G_i) < \rho n^2/2$. A path step changes at most $n - 2$ edges, and hence

$$\frac{\rho n^2}{2} - n \leq d_{\text{sp}}(G_i) < \frac{\rho n^2}{2}. \quad (8.4)$$

In particular, G_i lies inside the radius- ρn^2 neighbourhood, so lemma 5.2 applies. Combining (5.7) with (8.3) gives

$$\begin{aligned} \frac{d_{\text{sp}}(G_i)}{n^2} &\leq 18\eta + \sqrt{\frac{4\eta}{3}} + \frac{1}{n} \\ &< 18 \cdot 10^{-30} + 2 \cdot 10^{-15} + 10^{-32} < \frac{\rho}{4}. \end{aligned} \quad (8.5)$$

But $\rho/2 - 1/n > \rho/4$, so (8.4) and (8.5) are incompatible. This proves the theorem. $\square$

9. PROOF OF THE MAIN THEOREM

We now combine the near and far regimes. Recall the definition of N in (1.3).

Theorem 9.1. *Every chordal graph G on $n \geq N$ vertices satisfies*

$$\text{cp}_{\leq 4}(G) \leq M(n).$$

Proof. If (8.1) holds, then theorem 8.1 places G within the radius- ρn^2 neighbourhood. Equation (5.8) gives the stronger bound $\text{cp}_{\leq 3}(G) \leq M(n)$.

Otherwise, apply theorem 2.3 with $\varepsilon = \eta/2$. Using (2.4),

$$\begin{aligned} \text{cp}_{\leq 4}(G) &= \Phi(G) + W_4^*(G) - W_4(G) \\ &< \frac{n^2}{6} - \eta n^2 + \frac{\eta n^2}{2} < \frac{n^2}{6} \leq M(n), \end{aligned}$$

where the final inequality is valid for the present range of n . This completes both cases. $\square$

It remains to match the upper bound. We record the standard complete-split calculation because it also explains the linear term in $M(n)$.

Lemma 9.2. *If $\ell \geq \chi'(K_k)$, then*

$$\text{cp}(S_{k,\ell}) = k\ell - \binom{k}{2}. \quad (9.1)$$

Proof. Properly edge-colour the clique K_k using at most ℓ colours, identified with distinct vertices of the independent side. For every root edge xy , use the triangle consisting of x, y and the vertex corresponding to the colour of xy . Properness makes these triangles edge-disjoint. Cover every unused spoke by a K_2 . The resulting partition has

$$\binom{k}{2} + \left(k\ell - 2\binom{k}{2} \right) = k\ell - \binom{k}{2}$$

parts.

For the reverse inequality, assign weight -1 to each root edge and weight $+1$ to each spoke. Every clique in a complete-split graph has total edge weight at most 1: a clique contains at most one independent-side vertex, and, if it contains r root vertices and one outside vertex, its weight is $r - \binom{r}{2} \leq 1$. Its value is nonpositive when it contains only root vertices. Summing over any clique partition shows that the number of parts is at least the total edge weight $k\ell - \binom{k}{2}$. $\square$

Choose an integer k nearest to $(2n+1)/6$ and put $\ell = n - k$. Then $\ell \geq k \geq \chi'(K_k)$, using Vizing's theorem, and a check of the three residue classes of n modulo 3 gives

$$k(n-k) - \binom{k}{2} = \left\lfloor \frac{n(n+1)}{6} \right\rfloor = M(n). \quad (9.2)$$

Together with theorem 9.1, this proves theorem 1.1.

For completeness, let $C = N$. If $0 < n < N$, the partition into individual edges gives

$$\text{cp}(G) \leq \binom{n}{2} \leq \frac{n^2}{6} + Nn.$$

For $n \geq N$, theorem 9.1 and $M(n) \leq n^2/6 + n/6$ give the same conclusion. The empty graph is immediate. This proves corollary 1.2.

10. VERIFICATION AND SCOPE

The proof invokes three named non-elementary theorem inputs: theorems 2.1 to 2.3. In particular, it does not assume a linear bound for the general integral–fractional packing gap. The transfer theorem is invoked only in the far regime, where a fixed quadratic deficit pays for its $o(n^2)$ loss. In the near regime, lemma 3.1 constructs an actual edge partition into triangles and K_2 ’s.

The accompanying repository contains exact rational checks of the numerical inequalities and finite mixed-packing certificates. These computations are regression tests; none is a premise of the proof for arbitrary n .

The Lean companion checks the conditional upper-bound chain: terminal construction, root regularization, local stability, terminating single-vertex symmetrization, first entry, and near–far assembly. It also checks the signed-weight lower bound for complete-split witnesses and combines the two sides into the eventual exact extremal statement. Its external-input record has three interfaces corresponding to theorems 2.1 to 2.3; the packing-transfer interface packages attained rational primal and dual witnesses together with the published asymptotic gap. These are explicit theorem arguments rather than project axioms and have not themselves been reproved in Lean.

The formalization sometimes uses an equivalent proof architecture rather than mirroring the prose line by line. In particular, its terminal characterization uses minimal separators instead of clique trees, and path termination uses a finite lexicographic measure. The displayed auxiliary bound $L \leq n(n-1)$ is not needed by the final theorem and is not separately formalized. The rigidity addendum is supported by exact Python certificate replay but lies outside this Lean theorem perimeter. The repository records the precise trust ledger and the commands for reproducing the build and the fail-closed axiom audit.

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